Lectures 3 and 4

- rationalizable choice
- WARP
- optimal bundles: intuitive method

From Last Class

- a choice rule assigns, for any menu $A \subseteq X$, your non-empty set of choices $C(A) \subseteq A$
- a choice rule is *rationalizable* if there is a preference $\succsim$ that rationalizes it
  - i.e. such that *actual* choices are all those that are best, according to $\succsim$
  - notation: let $C_{\succsim}(A)$ be the set of $\succsim$-optimal options in A
- but, what conditions on choices guarantee (or rule out) existence of such a rationalizing preference?
- we'll present 2 necessary & sufficient conditions: IIA $=$ Sen's axiom α, and Sen's (consistency) axiom β
- then we'll state an equivalent axiom, WARP $=$ Weak Axiom of Revealed Preference

Independence of Irrelevant Alternatives

(Aka Sen's axiom α, or IIA):

Axiom IIA: If $x \in B \subseteq A$ and $x \in C(A)$, then $x \in C(B)$.

- In words, this says that if you choose x from some "big" set A, then you should also choose it from any subset of A containing it
- That is, the unchosen options in $A \backslash B$ should be irrelevant to how you evaluate options in B, so you should still be happy to choose x
- For example, it says:

If Simone Biles is the best gymnast in the world, then she must also be the best gymnast in the USA.

Proof: Rationalizability Implies IIA

- in words: a rational person would only choose x from A if it's most-preferred; but then it also beats anything on a subset B of the menu, so should be chosen here too
- formally: assume choices are rationalizable, so there exists $\succsim$ such that $C = C_{\succsim}$
- take $x \in B \subseteq A$ with $x \in C(A)$; we want to prove that $x \in C(B)$
- since $x \in C(A) = C_{\succsim}(A)$, x is $\succsim$-optimal in A
- but since $B \subseteq A$, this implies that x is $\succsim$-optimal in B
- so $x \in C_{\succsim}(B) = C(B)$

Notes

- previous slide proved that rationalizability $\Rightarrow$ IIA
- equivalently, the *contrapositive statement:* if choices do NOT satisfy IIA, they can NOT be rationalized
- IIA is *necessary* for rationalizability
- in a world where there's always only one thing you're willing to choose, it turns out that IIA is also sufficient for rationalizability
- but if choice sets may be multi-valued, the next slide shows that IIA is *not sufficient* for rationalizability

Observation II: IIA Does NOT Imply Rationalizability

- let $X = \{a, b, c\}$, and

$$\begin{aligned} C(\{a,b\}) &= \{a\}, C(\{a,c\}) = \{a,c\}, C(\{b,c\}) = \{c\} \\ C(X) &= \{a\}, \text{ and } C(\{x\}) = \{x\} \; \forall x = a, b, c \end{aligned}$$

- C satisfies IIA: a is chosen from X and also all subsets containing a, c is chosen from $\{a, c\}$ and also from $\{c\}$, and b is chosen only from $\{b\}$
- but if C were rationalizable, then $C(\{a, c\}) = \{a, c\}$ would indicate $a \sim c$; this is at odds with the implication from $C(X) = \{a\}$ that $a \succ c$

Sen's "Consistency" Axiom ("Beta")

- the problem with the example above is that it violates Sen's second axiom, "axiom β":

Sen's axiom β: If $\{x, y\} \subseteq C(B)$, $B \subseteq A$, and $x \in C(A)$, then $y \in C(A)$.

- For example: "If Simone Biles and Aly Raisman are tied for USA champions, and Simone is a world champion, then Aly Raisman must also be a world champion".

Proof: Rationalizability Implies Sen's Beta

- in words: the only way a rational person would choose both x and y from B is if $x \sim y$; but then if x is good enough to beat everything on an expanded menu A, so is equally good y
- formally: assume there is a preference $\succsim$ with $C = C_{\succsim}$, and that x and y are both chosen from $B \subseteq A$, with $x \in C(A)$
  - want to show that $y \in C(A)$
- since x and y are both chosen from B, and $C = C_{\succsim}$, we know x and y are both $\succsim$-optimal in B, thus $x \sim y$
- since $x \in C(A) = C_{\succsim}(A)$, we know x is $\succsim$-optimal in A
- but then by transitivity, since $x \sim y$, y is also $\succsim$-optimal in A
- so $y \in C_{\succsim}(A) = C(A)$

Observation: Beta Doesn't Imply Rationalizability

- reason: Sen's axiom β only has any "bite" if there is some set of alternatives from which the DM is willing to choose more than one alternative.
- Thus, take any choice rule in which $C(A)$ is a singleton for every set A, but which violates IIA
- this satisfies β, but is not rationalizable

A Theorem on Rationalizability

Theorem

Choices satisfy Sen's α and β axioms if and only if they are rationalizable

Proof Sketch

- we already proved one direction: rationalizability implies α and β must both hold
- for the converse, assume choices satisfy α and β
- to prove that they're rationalizable, construct a preference via

$$x \succsim y \text{ if } x \in C(\{x, y\})$$

- it now suffices to prove that actual choices are optimal according to this preference
- only way this could fail is if either (1) C chooses something that is NOT $\succsim$-optimal, or (2) C fails to choose something that IS $\succsim$-optimal

Proof Sketch

- can show that α rules out (1) and β rules out (2)
- for (1), if $x \in C(A)$ but x is not $\succsim$-optimal in A, there must be another $y \in A$ with $y \succ x$, i.e. $\{y\} = C(\{x, y\})$; this (choosing x from A but not its subset $\{x, y\}$) violates IIA
- for (2), precept will likely do this more slowly, but:
    - if x is $\succsim$-optimal in A but not chosen, then some $z \in C(A)$; but then by IIA, z is chosen from $\{x, z\}$ so is as good as x; but also $x \succsim z$ since premise is that x is $\succsim$-optimal in A. Thus $x \sim z$, so x, z are both chosen from $\{x, z\}$, so we violate β if z but not x is chosen from A.

Revealed Preference

- if a consumer chooses an option x when y was also feasible, we say that x is *directly revealed (weakly) preferred to y*
- more strongly, if x is chosen when y is feasible, but y is not, we say that x is directly revealed strictly preferred to y
- WARP: if you directly reveal (in one situation) that $x \succsim y$, then you can never reveal (in another situation) that $y \succ x$

Relationship Between Axioms

Proposition

A choice rule C satisfies WARP iff it satisfies Sen's $\alpha + \beta$ axioms.

Proposition

A choice rule C satisfies WARP if and only if it is rationalizable

Application: WARP and Classical Demand Theory

- in classical consumer demand theory, we assume that consumers are trying to find the best (utility-maximizing) choice in their *budget set*
- if a consumer can potentially spend his money on n goods, with wealth m and prices $p = (p_1, p_2, ..., p_n)$, then his *budget set* is the set of all affordable bundles, i.e. with $\sum_{i=1}^{n} p_i x_i \leq m$
- later today and next class, we'll learn how to actually solve this problem
- but for now, we'll specialize WARP to see what choices would immediately reveal irrational behavior

WARP and Classical Demand Theory

- in this setting, you directly reveal a (weak) preference for x over y if you choose x when y is also affordable
- and you directly reveal a **strict** preference for x over y if you choose x when it costs **more** than y
- so WARP says that if ever you buy x when y is also affordable, then you can never buy y when it costs MORE than x
- formally, WARP: if you choose x at prices p when $p \cdot x \geq p \cdot y$, then at any price p' where you choose y, it must be that $p' \cdot y \leq p' \cdot x$
- so you *violate* WARP if you choose x at prices p where $p \cdot x \geq p \cdot y$ [suggesting $x \succsim y$], and then choose y at prices p' where $p' \cdot y > p' \cdot x$ [suggesting $y \succ x$]

Summary Spin

- you *violate* WARP if:
  - at one set of prices, you buy bundle x when x is weakly more expensive than y (suggesting $x \succsim y$)
  - AND at another set of prices, you buy y when y is strictly more expensive than x (suggesting $y \succ x$)

Example 1

- On Monday, Homer Simpson buys 3 donuts and 0 coffees, when donuts cost \$1 and coffee costs \$1.50.
- On Tuesday, he buys 1 donut and 1 coffee.
- For what Tuesday prices does this violate WARP?

Example 1: Solution (Edited This Slide)

- the two bundles in play here are $M = (3,0)$ (the Monday bundle) and $T = (1,1)$ (the Tuesday bundle)
- Homer violates WARP if he reveals a preference flip: $M \succsim T$ and $T \succsim M$ with one strict
  - ie $M \succsim T$ and $T \succ M$, or $M \succ T$ and $T \succsim M$, or $M \succ T$ and $T \succ M$
- i.e., if one day he chooses M when it costs more [than T would have], and another day chooses T when it costs more [than M would have], with at least one "more" strict
- **note:** it's only safe to infer $x \succsim y$ if,
  <u>in the situation where you actually choose x</u>, it's the weakly more expensive option
- so let's see what preferences we can tease out here

Example 1: Monday Behavior

- he actually chooses M=(3,0) = (3 donuts, 0 coffees), when donuts cost \$1 and coffee costs \$1.50
- the chosen bundle (3,0) costs $3(\$1) + 0(\$1.50) = \$3$
- at these **same** Monday prices, the unchosen Tuesday bundle (1,1) *would have cost* $1(\$1) + 1(\$1.50) = \$2.50$
- <u>this reveals $M \succ T$</u>: he chooses M=(3,0) in a scenario where he pays extra for it, compared to T=(1,1)

Example 1: Tuesday Behavior

- so Homer violates WARP if he does something else revealing the opposite, $T \succsim M$
- i.e. in another situation, he picks T when **it** is the (weakly) pricier option
- i.e., if the chosen bundle T=(1,1) costs weakly more than M=(3,0) would have, at the Tuesday prices:

$$(1,1) \cdot (p_D^T, p_C^T) \geq (3,0) \cdot (p_D^T, p_C^T)$$

- i.e. if $p_{\text{donut}}^{\text{Tues}} + p_{\text{coffee}}^{\text{Tues}} \geq 3p_{\text{donut}}^{\text{Tues}}$
- simplifying, Homer violates WARP if on Tuesday, coffee costs more than twice as much as a donut

Optimal Bundles

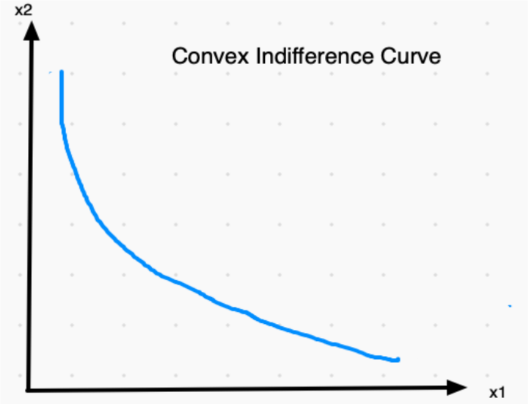
- we are now going to learn a few methods for actually finding the best affordable bundle
- for now, specialize to the 2-good case
- consider a consumer who gets utility $u(x_1, x_2)$ from the bundle (x_1, x_2), who wants to choose the best affordable bundle
- assume prices p_1 and p_2 for goods 1 and 2, and wealth m
- "affordable" $\iff$ "doesn't cost more than you have":

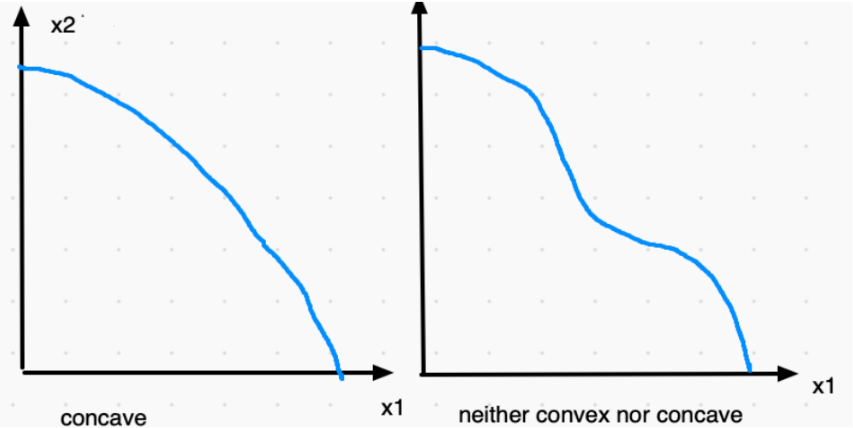
$$p_1 x_1 + p_2 x_2 \leq m$$

- comment: in practice, the best bundle will satisfy this with equality
  - if you don't care about money per se, then it's best to spend it on stuff you *do* value

Preamble: Convexity

- *convex preferences* are those where the indifference curve has a convex (up) shape, i.e. flattens out as you move down and to the right
- mathematically, this happens if $MU_1 \,/\, MU_2$ (the steepness of the indifference curve) falls as $x_2 \downarrow$ and $x_1 \uparrow$
  - this is an implication of *diminishing returns*, which is usually true: the more you have of a good, the less you value it
  - for then as $x_1 \uparrow$, $MU_1 \downarrow$; and as $x_2 \downarrow$, $MU_2 \uparrow$
- an equivalent check is the "straight line test": for any 2 points on an indifference curve, does the line segment connecting them lie above the indifference curve? Yes –> convex. No –> Not convex.


x2
Convex Indifference Curve
x1


x2
x1
concave
x1
neither convex nor concave

Preamble: Convexity

- "diminishing returns" suggests that extreme bundles are bad, as it would be better to buy a more balanced bundle
- this is why we care about convexity (it relates to the second-order condition for utility maximization)
- if preferences are convex, then you are generally looking for an *interior* optimal bundle
- if preferences are concave, then you are looking for a *corner solution:* all money on one of the two goods

Optimal Bundles with Convex Preferences

- the goal: choose (x_1, x_2) to maximize $u(x_1, x_2)$, subject to the *budget* and *non-negativity* constraints:
  - budget: $p_1 x_1 + p_2 x_2 \leq m$
  - non-negativity: $x_1 \geq 0$ and $x_2 \geq 0$
- we'll focus on 2 methods to solve this:
  1. Intuitive "Bang-Per-Buck" Solution
  2. Lagrangian

Intuitive Solution: The Recipe

- solve two equations in two unknowns, x_1 and x_2
- equation #1: spend all your money, i.e. $p_1x_1 + p_2x_2 = m$
- equation #2: allocate your money optimally, by (**if possible**) equating *bang-per-buck*:

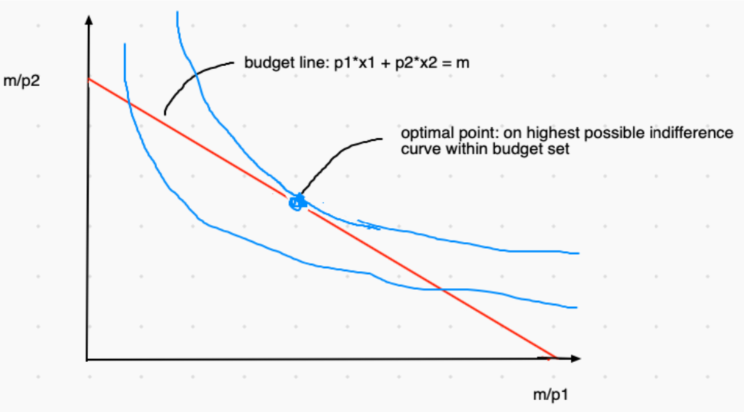
$$\frac{MU_1}{p_1} = \frac{MU_2}{p_2}$$

  - I've called this the "optimal spending condition" or "bang-per-buck" condition in upcoming slides
  - in practice, the more helpful rearrangement is usually $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$
- caveat: if any feasible (i.e. affordable and non-negative) bundle entails $\frac{MU_1}{p_1} > \frac{MU_2}{p_2}$, choose a corner: all \$ on good 1
- caveat: if any feasible bundle entails $\frac{MU_2}{p_2} > \frac{MU_1}{p_1}$, choose a corner: all \$ on good 2

Intuitive Solution: The Intuition

- to make yourself as happy as possible, spend all your money, and allocate it in the smartest way possible
- if one good has more bang-per-buck than the other, this is a nudge that you could do better by reallocating more $ to this good
  - so this isn't optimal, unless you're already at the corner where all your $ is on the high-bang-per-buck good
- but if the two goods have equal bang-per-buck, you can't do any better by reallocating $ from one to the other

Intuitive Solution: The Picture



Intuitive Approach: Example 1

Suppose Pete's utility from B burgers and A apples is $u(B, A) = B^{\frac{1}{2}} A^{\frac{1}{2}}$. If each burger costs p_B, each apple costs p_A, and he has $\$m$ available, what should he buy?

Example 1: Solution

- we know that the optimal bundle (B, A) satisfies two equations: it's on the budget line, so

$$p_B B + p_A A = m \tag{1}$$

- and it satisfies the optimal spending condition, assuming preferences are convex:

$$\frac{MU_B}{MU_A} = \frac{p_B}{p_A}$$

Example 1: Solution

- first, let's calculate marginal utility
- here $MU_B = \frac{1}{2}A^{\frac{1}{2}}B^{-\frac{1}{2}}$, and $MU_A = \frac{1}{2}A^{-\frac{1}{2}}B^{\frac{1}{2}}$
- note that MU_B falls in B, and MU_A falls in A, so **preferences are convex**
- so the optimal spending condition is:

$$\begin{aligned} \frac{p_B}{p_A} &= \frac{MU_B}{MU_A} = \frac{\frac{1}{2}A^{\frac{1}{2}}B^{-\frac{1}{2}}}{\frac{1}{2}A^{-\frac{1}{2}}B^{\frac{1}{2}}} = \frac{A}{B} \\ \Rightarrow & \quad p_B B = p_A A \end{aligned} \tag{2a}$$

- now, plug this into the budget line to get the solution: spend half your income on each good
- that is, $p_B B = p_A A = \frac{m}{2}$; optimal bundle itself $(A, B) = \left(\frac{m}{2p_A}, \frac{m}{2p_B}\right)$

Side Note: Altering the Utility Function

- we say that two utility functions u and v *represent the same preferences* if they rank bundles the same way, i.e.

$$A \succsim B \Leftrightarrow u(A) \geq u(B) \Leftrightarrow v(A) \geq v(B)$$

- this is guaranteed if v is an *increasing transformation of u*
    - eg $v = 2u$, or $v = \log u$, or $v = u^2$ (assuming $u \geq 0$)
- so you are free to take an increasing transformation, which sometimes can simplify the algebra
    - it won't change the preferences
    - so it won't change your answer

Illustration: Altering the Utility Function

- suppose that instead of using utility $B^{\frac{1}{2}}A^{\frac{1}{2}}$ in above example, we square it: $u(A, B) = BA$
- then $MU_B = A$, and $MU_A = B$, so the new optimal spending condition is

$$\begin{aligned} \frac{MU_B}{MU_A} &= \frac{p_B}{p_A} \Leftrightarrow \frac{A}{B} = \frac{p_B}{p_A} \\ &\Rightarrow p_A A = p_B B \end{aligned}$$

- exactly as we had before!, just with slightly prettier algebra

Example 1, Generalized

- we found that with utility $u(x_1, x_2) = x_1^{\frac{1}{2}} x_2^{\frac{1}{2}}$, it is optimal to spend half your wealth on each good, ie $p_1 x_1 = p_2 x_2 = \frac{m}{2}$
- what about $u(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}$?
- conjecture?

Example 1 Generalized

- let's rescale the utility function by taking its natural log, to obtain

$$U(x_1, x_2) = \alpha \ln x_1 + (1-\alpha) \ln x_2$$

- we know that the optimal bundle solves two equations:
  - spend all your \$: $p_1 x_1 + p_2 x_2 = m$ (1)
  - spend \$ optimally, ie equate bang-per-buck: $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$ (2)
- here, (2) says

$$\frac{\alpha / x_1}{(1-\alpha)/x_2} = \frac{p_1}{p_2} \Rightarrow \alpha p_2 x_2 = (1-\alpha) p_1 x_1$$

- plugging this into (1) and solving gives $p_1 x_1 = \alpha m$, $p_2 x_2 = (1-\alpha) m$

“Magic Rule" for Cobb-Douglas Preferences

- Cobb-Douglas preferences are ones that can be represented by $u(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}$
- and we just derived the “magic rule”: the exponents tell you the income share that optimally goes to each good!

$$\text{optimal: } x_1 = \frac{\alpha m}{p_1}, x_2 = \frac{(1-\alpha)m}{p_2}$$

- can you use this to figure out what to buy if $u(x_1, x_2) = x_1 x_2^2$, and $m = 15$ and $p_1 = p_2 = 1$?

Intuitive Method: Example 2

- now consider perfect substitutes: $u(x_1, x_2) = x_1 + x_2$
- well here, $MU_1 = MU_2 = 1$ at *every* point, so it may not be possible to equate the bang-per-bucks
- solution:
  - if $p_1 < p_2$, then $\frac{MU_1}{p_1} > \frac{MU_2}{p_2}$ at EVERY feasible point: this means good 1 ALWAYS has more bang-per-buck, so spend all your money on it
  - similarly if $p_2 < p_1$, spend everything on good 2
  - idea: if you value pens and pencils equally, then you'll just buy whichever is cheaper
  - if $p_1 = p_2$, ANY affordable bundle is optimal

Summary and Intuition for Example 2

- so the optimal bundle is anything on the budget line (with $p_1x_1 + p_2x_2 = m$) if $p_1 = p_2$, and otherwise

$$(x_1, x_2) = \begin{cases} \left(\frac{m}{p_1}, 0\right) & \text{if } p_1 < p_2 \\ \left(0, \frac{m}{p_2}\right) & \text{if } p_1 > p_2 \end{cases}$$

- with a bit of thought, you could have jumped to this
- imagine you have no preference between pens vs pencils, but want as many things to write with as possible
- how should you spend your money if pencils are cheaper? What if pens are cheaper?

Intuitive Method: Example 3

- now consider perfect complements, with $u(L, R) = \min\{L, R\}$
- as usual, we want a point on the budget line, i.e. with $p_L L + p_R R = m$
- but the usual optimal spending condition ($\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$) obviously doesn't work here, what do you think replaces it?
- I'll fill this in after class

Intuitive Method: Example 4

- now take $u(x_1, x_2) = x_1 + 2x_2\sqrt{x_1} + x_2^2$, with $p_1 = 1, p_2 = 4$, income m
- here, if you use this utility function directly, you get

$$\frac{MU_1}{MU_2} = \frac{1 + \frac{2x_2}{2\sqrt{x_1}}}{2\sqrt{x_1} + 2x_2} = \frac{1}{2\sqrt{x_1}}$$

(or notice $u(x_1, x_2) = (\sqrt{x_1} + x_2)^2$, and work with its square root, which is a monotone transformation)

- so we want a point that's on the budget line, i.e.

$$x_1 + 4x_2 = m \qquad (1)$$

- and that equates bang-per-buck across goods (if possible), i.e.

$$\frac{1}{2\sqrt{x_1}} = \frac{1}{4} \Rightarrow \sqrt{x_1} = 2 \Rightarrow x_1 = 4 \qquad (2)$$

Intuitive Method: Example 4 Cont'd

- if $m \geq 4$, this all works nicely: we already have the optimal amount of good 1, from (2), and for the optimal x_2, plug (2) into (1) to get

$$4 + 4x_2 = m \Rightarrow x_2 = \frac{m-4}{4}$$

- possible problem: what if $m < 4$?
- here, your math told you $x_1 = 4$ and $x_2 = \frac{m-4}{4}$, which is negative (not allowed)
- quick-and-dirty solution: buy zero of the good where your math gives you a negative quantity
- so optimal bundle is:

$$x_2 = 0, x_1 = \frac{m}{p_1} = m$$

Intuitive Method: Example 4 Cont'd

- logic behind quick-and-dirty solution:
  - feasibility implies $x_1 \leq \frac{m}{p_1}$
  - so if $p_1 = 1$ and $m < 4$, then feasibility $\Rightarrow x_1 < 4$, and so

$$\begin{aligned} \frac{MU_1}{MU_2} &= \frac{1}{2\sqrt{x_1}} > \frac{1}{4} \\ \text{while } \frac{p_1}{p_2} &= \frac{1}{4} \end{aligned}$$

- Thus at income levels $m < 4$, good 1 has more bang-per-buck at any feasible point
- In other words, we're in the caveat case where good 1 always has more bang-per-buck, so spend all $ on it

Example 5

- assume $u(x_1, x_2) = x_1^2 + x_2$ and $p_1 = 2$ and $p_2 = 1$
- what would you buy at $m = 1$? what about $m = 5$?
- solution: here $MU_1 = 2x_1$ (*increasing* not diminishing returns!) and $MU_2 = 1$
- so $\frac{MU_1}{MU_2} = 2x_1$, which **rises** as $x_1 \uparrow$ and $x_2 \downarrow$
- these are addictive concave preferences!
- so a corner solution is best: either $\left(\frac{m}{p_1}, 0\right)$ or $\left(0, \frac{m}{p_2}\right)$
- at $m = 1$, either $\left(\frac{1}{2}, 0\right)$ or $(0, 1)$: the latter is best (utility 1 rather than $\frac{1}{4}$)
- at $m = 5$, either $\left(\frac{5}{2}, 0\right)$ or $(0, 5)$: the former is best (utility 6.25 instead of 5)

Space for Pictures